PSCI 2075: Quantitative Research Methods

University of Colorado Boulder, Fall 2026

> **Note**
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> **Syllabus Version:** This document was last updated on **August 23, 2026**. This syllabus is a living document and may change during the semester — always refer to the most up-to-date version on Canvas (PDF version) rather than a saved or printed copy. Please see the Syllabus Change Log for information on any changes made to this syllabus throughout the semester.

Course Overview

Course & Contact Information

Lecture: Monday & Wednesday, 1:25–2:15 PM, Gold Biosciences Building, A2B70

Instructor

Dr. Fabian Neuner

Email Fabian.Neuner@colorado.edu
Office Hours Wednesdays 10:00 AM–12:00 PM, and by appointment
Location KTCH 114F

Teaching Assistants

Name	Email	Office Hours	Location
Claudette Medina	Claudette.Medina@colorado.edu	Mondays 11:00 AM–1:00 PM	KTCH 382
Donald Caviness	Donald.Caviness@colorado.edu	Thursdays 8:50–10:50 AM	KTCH 382

Name	Email	Office Hours	Location
Romulo Hannig	Romulo.Hannig@colorado.edu	Thursdays 10:00–12:00 AM	KTCH 412
Steven Van De Laarschot	Steven.VanDelaarschot@colorado.edu	Wednesdays 10:00 AM–12:00 PM	KTCH 234

Methods Lab Coordinator

Tania Massicotte

Email tania.massicotte@colorado.edu

Office Hours Beginning August 30, 2026: Mondays–Fridays 8:00–11:00 AM, and Tuesdays & Thursdays 5:00–7:30 PM. Before then: nightly 7:00–10:00 PM (including weekends).

Location Zoom (link provided upon booking)

Appointments Book a 20-minute slot via Calendly at least one day in advance. Need more time? Book consecutive slots, being mindful of other students' availability. Your confirmation email includes a link to cancel if your plans change.

Class Meeting Times

Section	Day	Time	Location	Instructor
Lec. (010)	M/W	1:25–2:15 PM	GOLD A2B70	Dr. Neuner
Rec. (011)	M	11:15 AM–12:05 PM	CASE E224	Prof. Hannig
Rec. (012)	M	2:30–3:20 PM	MCOL E155	Prof. Medina
Rec. (013)	T	10:10–11:00 AM	MCOL E155	Prof. Medina
Rec. (014)	W	9:05–9:55 AM	CLRE 209	Prof. Van De Laarschot
Rec. (015)	W	4:40–5:30 PM	MUEN E064	Prof. Hannig
Rec. (016)	TH	8:00–8:50 AM	KTCH 1B60	Prof. Caviness
Rec. (017)	T	4:40–5:30 PM	CASE E230	Prof. Caviness
Rec. (018)	F	1:25–2:15 PM	GOLD A2B07	Prof. Van De Laarschot

Course Description

Data are all around us. Quantitative data nowadays appear every single day in the media, in politics, the workplace, and even in our own lives. This course will introduce you to quantitative research methods and the aim is to turn you into a better consumer—and even a producer—of analyses using quantitative data. These skills are becoming increasingly more important in both public and private sector careers. Moreover, AI is quickly reshaping not only the analysis of quantitative data, but the job market landscape and society itself. The skills learned in this course will also be needed in future courses you may take in the political science department. The goal is to familiarize you with analyzing, presenting, and interpreting patterns in data. While the data used in this class will place an emphasis on the social sciences, the tools learned in this class are easily applied to many other settings.

Most weeks will consist of a lecture in class on Monday and a more interactive session on Wednesday in which we will work through coding in R together. Each of you are also assigned a recitation session, which is taught by one of the course TAs. What is covered in recitation differs by week; some may delve deeper into a topic we cover in lecture. Other times it may involve allowing time to work on assignments, or reviewing for exams and individual assignments.

Note that this is an involved, hands-on, rigorous course. Yet it is neither a math nor programming course. You will need to complete readings each week, complete any assignments, and stay on top of the course material. Students can usually expect to spend around 8-12 hours per week on classes/assignments/readings related to this course.

Goals of the Course

By the end of this course you should be able to:

- Be familiar with basic research design and statistical concepts
- Be a better and more engaged consumer of data
- Be able to manipulate and analyze data using R
- Be a producer of quantitative analyses
- Be aware of how careful AI use can assist with the above

Prerequisites

There are no prerequisites for this course.

Technological Requirements & Recommendations

Laptop Computer

Access to a laptop computer is required for this class. Laptops can be rented for the semester from the CU Boulder Bookstore.

R and RStudio

We will be using R and RStudio for coding and statistical analysis in this class. We will help you install these programs and no prior experience with these applications is necessary.

CUClickers

This class will use CUClickers for class participation and activities. We will not be using physical clickers and participation will only require the iClicker mobile app or web browser. Instructions for setting up your account will be provided in class and you should automatically be enrolled in the class on iClicker.

CU Secure Wifi

Please sign up for new CU Secure Wifi Network for seamless connectivity during class activities.

ChatGPT Edu

Students are eligible for ChatGPT Edu access and we encourage you to complete the training so that you can access this resource. See the AI section for details on how generative AI can (and cannot) be used in this class.

Textbook Information

In this class we will primarily use the following two **required** texts:

- Pollock & Edwards, *The Essentials of Political Analysis* (EPA), 7th ed.
- Pollock & Edwards, *An R Companion to Political Analysis* (RCPA), 3rd ed.

In addition, there is a **highly recommended** text that is used to supplement the above:

- Brown, *Statistics and Data Visualization Using R: The Art and Practice of Data Analysis* (2021, SAGE)

Our expectation is that you will have read the **required** texts prior to the relevant class sessions. We will also be using the **highly recommended** text and you are encouraged to read those for the relevant class sessions as well as using it as reference material throughout the course. If you ever feel confused about a topic after reading the EPA and or RCPA text, your best next move will be to read the relevant chapter in Brown for additional information.

Unless you opt out of the CU Book Access program, you should have access to these books through Canvas (click on "CU Book Access" at the bottom left). If you opt out, you will be expected to access the texts through the library or acquire copies of the books.

Syllabus Acknowledgments

This syllabus and the course itself draw on materials, assignments, and policies developed and refined by various previous PSCI 2075 instructors. Special thanks go to Professors Philips, Beard, and Parinandi for sharing course materials. The syllabus was refined, edited, and organized with the assistance of generative AI.

Assignments & Grading

In this course you will be assessed through a variety of ways and the below table provides an overview of all assessments and how much they contribute to your overall grade.

For all assignments that require a "submission", details will be provided on Canvas when the assignment becomes available. This will include the grading rubric used and the rules for AI use for that specific assignment.

Grade Breakdown

Component	Weight
Recitation Grade	10%
Attendance & Class Participation	10%
Individual Data Project	10%
Group Assignments	25%
Midterm Exam	20%
Final Exam	25%

Assignment Details

Please find more detailed information on all assessment methods used for this class below. Additional details will be provided in class, recitation, and on Canvas.

> **Note**
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> **All due dates listed below are tentative.** Official due dates will be confirmed on Canvas, which is always the authoritative source if a date here and a date on Canvas ever disagree. Any changes made to these tentative dates will be noted in the Syllabus Change Log.

Recitation Grade

Attendance in your recitation section is required and you are expected to actively participate in group activities, exercises, and discussions. Your TAs will track attendance and have discretion on how to award points for participation. You are allowed two unexcused absences for recitations. Your TA will explain any additional recitiation policies. You will be provided with update on your recitation participation grade halfway through the semester.

Attendance & Class Participation

Every class session you will receive base points for showing up and then you will be able to earn additional points based on participation in class activities and completion of mini quizzes administered during class using CUClicker. These points will be continually updated throughout the semester. Your three lowest scores will be dropped, meaning that you are effectively allowed three *"no questions asked"* absences from lecture.

Individual Data Project

Over the course of the semester, you will build and analyze your own original dataset, applying course concepts to a topic of your choosing. This project unfolds in stages, from an initial proposal through a final analysis. The assignment sequence will give you hands-on practice from developing a research idea, to collecting data, to interpreting statistical analyses. These are low-stakes assignments with each individual assignment contributing 2% to your grade.

The **tentative** schedule for these assignments is:

- Topic idea and "pre-registration" is tentatively due Tuesday, September 8
- Dataset construction is tentatively due Tuesday, September 22
- Data wrangling and Descriptive Statistics is tentatively due Tuesday, October 6
- Basic Data Analysis is tentatively due Tuesday, November 3
- Reflection and forward design is tentatively due Tuesday, December 1

Group Assignments

Working in small groups of 2-3 (or individually if you choose), you will complete a series of assignments. These assignments will contain a mix of tasks (such as coding, short answers, code debugging, analysis, interpretation) and build on one another over the course of the semester.

Each member of a group will have to submit their own version of the assignment on Canvas and each assignment will include a final question that will need to be completed individually.

There are three group assignments over the course of the semester, each worth an equal share of the 25% Group Assignments component of your grade (25% ÷ 3 8.33% per assignment).

The **tentative** schedule for these assignments is:

- Group Assignment 1 is tentatively due Thursday, October 1
- Group Assignment 2 is tentatively due Thursday, October 29
- Group Assignment 3 is tentatively due Thursday, November 19

Exams

Both exams will be in person, using blue/green books, will consist of a mix of multiple choice and short-answer questions, and must be completed on your own. Exams are cumulative, in that everything covered in the course up to that point is potentially on the exam.

- **Midterm Exam:** The midterm exam will take place **in person** during class session on Wednesday 10/14.
- **Final Exam:** The final exam will take place **in person** on Thursday, 12/10, 1:30–4:00 PM.

Grading Scale

We will be using the below scale to turn numerical grades into letter grades. Please note that we will round up a letter for grades ending in .5 or above. For example, an 89.5 will round to 90 and thus an A-. This rounding is automatic and will be reflected in the letter grade you see in Canvas.

Grade	Range
A	94–100
A-	90–93
B+	87–89
B	84–86
B-	80–83
C+	77–79

Grade	Range
C	74–76
C-	70–73
D+	67–69
D	64–66
D-	60–63
F	0–59

Grading Grievances

If you have questions about any grade you receive in this class, your first point of contact should be your TA who will be happy to review the assignment in question during office hours. If, after such a meeting, you wish to appeal a grade you should email both me and the TA a formal request for a regrade in which you explain your reasons for the request. Any such requests need to be submitted within two weeks of you receiving the grade.

A Note on "Points" in Canvas

Please note that we are using *"weighted group"* assignments in Canvas. This means that each assignment category is worth the amount of points listed in the Grade Breakdown above even if you see different point values in Canvas. This is to facilitate grading and to make rubrics/points more intuitive. For instance, each of the three group assignments is worth 8.3% toward your final grade but rather than scoring each assignment out of 8.3 points, they are scored out of 25 for convenience.

Extra Credit

Throughout the semester there may be opportunities for extra credit. Any extra credit opportunities will be announced through Canvas.

Policies

Late Work

Unless explicitly noted, all assignments will be submitted through Canvas. Grades for late assignments are reduced 10 points for every day late (e.g., what would be graded as an A- is now a B-). Between four and ten calendar days following the due date, students can receive a maximum score of 60% for the assignment. Unless a specific accommodation was provided, assignments cannot be submitted for credit after ten days following the original due date.

Attendance and Make-Up Work Policy

Please note that exams, quizzes, and other participation points cannot be made up without a university-excused absence.

If you have to miss class sessions or exams or are in need of an extension due to health-related reasons or religious obligations please contact both Professor Neuner and your TA — at least two weeks in advance for religious obligations, or as soon as you are able for health-related reasons. We will work with you to coordinate make-up opportunities and faciliate extensions as needed.

Contact with Instructors

Please contact your instructors through Canvas. This allows us to keep a clear record of communications and ensures nothing gets lost. You can usually expect a response to your messages within 24 hours on weekdays and can follow up if you have not received a response within 48 hours. Please ensure that all communications are clear, professional, and that you include code/screenshots when asking about issues with R.

Course Alerts

This course participates in a campus effort to alert students early if there are concerns about grades, attendance, or performance. The goal is to give timely feedback and connect students with support resources.

Schedule & Readings

Weekly Schedule

Below is the weekly schedule for this class. In most weeks there will be a lecture on the Monday while on Wednesday we will work through code in R together.

Reading key

EPA = Essentials of Political Analysis (7th ed.) · **RCPA** = An R Companion to Political Analysis (3rd ed.) · **Brown** = Statistics and Data Visualization Using R (2021) Please note that the chapter numbers below refer to the 7th edition of EPA. The textbook was heavily revised since the 6th edition so please reach out for guidance if not accessing the 7th edition.

> **Note**
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> Please note that the **Assessments** column includes the dates for your midterm and final, which are confirmed dates. We have also included the **tentative** dates for all other assignments. See the Assignment Details section for more information on the assignments and how they contribute to your grade.

Week	Dates	Topic	Readings	Assessments
1	Mon 8/24, Wed 8/26	Introduction and Research Design Overview	Syllabus; EPA Introduction (Wednesday)	—
2	Mon 8/31, Wed 9/2	Measurement and Getting Started with R	EPA 1 · RCPA Intro & Ch. 1 · *Brown 1*	—
3	Wed 9/9; **No class on Monday (Labor Day)**	Measuring and Describing Data	EPA 2 · RCPA 2 · *Brown 3 & 4*	Individual 1 · Tue 9/8
4	Mon 9/14, Wed 9/16	Transforming Variables	EPA 3 · RCPA 3 · *Brown 6*	—
5	Mon 9/21, Wed 9/23	Comparisons	EPA 4 · RCPA 4 · *Brown 10*	Individual 2 · Tue 9/22
6	Mon 9/28, Wed 9/30	Graphing Relationships	EPA 5 · RCPA 5 · *Brown 5*	Group 1 · Thu 10/1
7	Mon 10/5, Wed 10/7	Controlled Comparisons	EPA 7 · RCPA 7 · *Brown 11*	Individual 3 · Tue 10/6
8	Mon 10/12, Wed 10/14	Midterm Exam Week	Monday: Review & Catchup	**Midterm Exam** · Wed 10/14, in person
9	Mon 10/19, Wed 10/21	Statistical Inference & Hypothesis Testing	EPA 8 & 9 · RCPA 8 & 9 · *Brown 8 & 9*	—
10	Mon 10/26, Wed 10/28	Correlation and Bivariate Regression	EPA 11 · RCPA 11 · *Brown 12*	Group 2 · Thu 10/29
11	Mon 11/2, Wed 11/4	Multiple Regression	EPA 12 · RCPA 12 · *Brown 13 & 14*	Individual 4 · Tue 11/3
12	Mon 11/9, Wed 11/11	More Regression (& Diagnostics)	EPA 13 · RCPA 13 · *Brown 15 & 16*	—
13	Mon 11/16, Wed 11/18	Logistic Regression	EPA 14 · RCPA 14 · *Brown 17*	Group 3 · Thu 11/19

Week	Dates	Topic	Readings	Assessments
14	Mon 11/23, Wed 11/25	**No classes — Fall Break**	—	—
15	Mon 11/30, Wed 12/2, **Fri 12/4***	Causality, Experiments, & Course Review	EPA 6 · RCPA 6	Individual 5 · Tue 12/1
16	Thu 12/10, 1:30–4:00 PM	Final Exam Period	—	**Final Exam** · Thu 12/10, 1:30–4:00 PM, in person

* Friday, December 4th follows a **Monday schedule**. This is a university-wide "double Monday" adjustment which means that we will also be holding a class session on Friday that week.

University-Mandated Statements

Honor Code

All students enrolled in a University of Colorado Boulder course are responsible for knowing and adhering to the Honor Code. Violations of the Honor Code may include but are not limited to: plagiarism (including use of paper writing services or technology [such as essay bots]), cheating, fabrication, lying, bribery, threat, unauthorized access to academic materials, clicker fraud, submitting the same or similar work in more than one course without permission from all course instructors involved, and aiding academic dishonesty. Understanding the course's syllabus is a vital part of adhering to the Honor Code.

All incidents of academic misconduct will be reported to Student Conduct & Conflict Resolution: StudentConduct@colorado.edu. Students found responsible for violating the Honor Code will be assigned resolution outcomes from Student Conduct & Conflict Resolution and will be subject to academic sanctions from the faculty member. Visit Honor Code for more information on the academic integrity policy.

Accommodation for Disabilities, Temporary Medical Conditions, and Medical Isolation

If you qualify for accommodations because of a disability, please submit your accommodation letter from Disability Services to your faculty member in a timely manner so that your needs can be addressed. Disability Services determines accommodations based on documented disabilities in the academic environment. Information on requesting accommodations is located on the Disability Services website. Contact Disability Services at 303-492-8671 or DSinfo@colorado.edu

for further assistance. If you have a temporary medical condition, see Temporary Medical Conditions on the Disability Services website.

If you have a temporary illness, injury or required medical isolation for which you require adjustment, you must communicate this to the instructor and/or your TA. You do not need to disclose the nature of your illness.

Accommodation for Religious Obligations

Instructional faculty members must make every reasonable effort to accommodate all students who have conflicts with scheduled exams, assignment deadlines or required attendance due to a religious observance. Whenever possible, students must notify the instructional faculty member at least two weeks in advance of the expected exam, assignment deadline, or attendance conflict to request an accommodation for religious observance. If the start date of the course is less than two weeks before the date of requested accommodation, students must notify the instructional faculty member on the start date of the course. See the Student Academic Accommodations for Religious Observances Policy for more information.

In this class, please contact Professor Neuner and your TA as soon as you become aware of the need for an accommodation for religious obligations so that we can facilitate the accommodation.

Student Names and Pronouns

CU Boulder recognizes that students' legal information does not always align with how they identify. If you wish to have a name other than your legal name appear on your instructors' class rosters and in Canvas, or if you wish to choose pronouns to appear on your instructors' class rosters and in Canvas, visit the Registrar's website for instructions on how to change your personal information in university systems.

Classroom Behavior

Students and faculty are responsible for maintaining an appropriate learning environment in all instructional settings, whether in person, remote, or online. Failure to adhere to such behavioral standards may be subject to discipline. Professional courtesy and sensitivity are especially important with respect to individuals and topics dealing with race, color, national origin, sex, pregnancy, age, disability, creed, religion, sexual orientation, gender identity, gender expression, veteran status, marital status, political affiliation, or political philosophy.

Additional classroom behavior information

- Student Classroom and Course-Related Behavior Policy.

- Student Code of Conduct.
- Office of Institutional Equity and Compliance.

Sexual Misconduct, Discrimination, Harassment and/or Related Retaliation

CU Boulder is committed to fostering a productive and welcoming learning, working, and living environment. University policy prohibits protected-class discrimination and harassment, sexual misconduct (harassment, exploitation, and assault), intimate partner abuse (dating or domestic violence), stalking, and related retaliation by or against members of our community on- or off-campus. Denial of an approved accommodation for disability, religious observance, or pregnancy or pregnancy related medical conditions may be discriminatory.

The Office of Institutional Equity and Compliance (OIEC) addresses these concerns, and individuals who have been subjected to misconduct can contact OIEC at 303-492-2127 or email OIEC@colorado.edu. Information about university policies, OIEC reporting options, and OIEC support resources including confidential services can be found on the OIEC website.

Faculty and graduate instructors are required to inform OIEC when someone discloses misconduct regardless of when or where it occurred. This is to ensure that those impacted receive outreach from OIEC about resolution options and support resources. To learn more about reporting and support options for a variety of concerns, visit the Don't Ignore It page.

Mental Health and Wellness

The University of Colorado Boulder is committed to supporting students' mental health and overall wellbeing. If personal, academic, or emotional challenges are affecting your wellbeing or success, Counseling and Psychiatric Services (CAPS), is here to help. CAPS offers counseling, referrals, psychiatric care, crisis support, and much more. Visit CAPS in the C4C, or call (303) 492-2277, 24/7.

Acceptable Use of AI in This Class

Generative artificial intelligence tools—software that reproduces text, images, computer code, audio, video, and other content—have become widely available. Well-known examples include ChatGPT for text and DALL • E for images. This statement governs all such tools, including those released during our semester together. Keep in mind that the goal of gen AI tools is to reproduce content that seems to have been produced by a human, not to produce accurate or reliable content; therefore, relying on a gen AI tool may result in your submission of inaccurate content. It is your responsibility—not the tool's—to assure the quality, integrity, and accuracy of work you submit in any college course.

If gen AI tool use is suspected in completing assignments for this course in ways not explicitly authorized, I will follow up with you. I may contact the Office of Student Conduct & Conflict Resolution to report suspected Honor Code violations. In addition, you must be wary of unintentional plagiarism or data fabrication. Please act with integrity, for the sake of both your personal character and your academic record.

The use of generative AI is strictly prohibited for exams and quizzes. However, some use of AI is permitted in this class and even encouraged. For instance, you can use AI for assistance coding, for brainstorming ideas, and for debugging code. We will provide specific guidelines on AI permissions for each assignment and will also discuss how to integrate AI into your workflow during class and recitation. If you are ever in doubt about whether AI use is acceptable, please contact your instructors.

Syllabus Change Log

This document was last updated on August 23, 2026 (official start of the semester version).